MIL-E-1/1460A
6 June 1975
SUPERSEDING
MIL-E-1/1460 (Sig C)
8 June 1962

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5875

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, rf sharp cutoff
Outline --- 2-1 (EIA)
Base --- Flat press
Envelope --- TZX3 (see note 1)
Cathode --- Coated filament

Base connections:

| Pin No. --- | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Element --- | a (red dot) | g2 | -f sd, g3 (see note 2) | g1 | +f g3 (see note 2) |

ABSOLUTE RATINGS:

| Parameter: | Ef | Eb | Ec2 | Ec1 | Ik | Alt |
|---|---|---|---|---|---|---|
| Unit: | Vdc | Vdc | Vdc | Vdc | mAdc | ft |
| Maximum: | 1.375 | 100 | 100 | --- | 7.0 | Note 3 |
| Minimum: | 1.125 | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | 1.25 | 90 | 90 | 0 | --- | --- |

GENERAL:

Preproduction sample approval - Required (see note 4)

Ⓐ denotes changes



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Preproduction sample approval inspection | | | | | | | | |
| 1202 | Filament burnout | - | | --- | --- | --- | --- | --- | --- |
| 1331 | Direct-interelectrode capacitance | - | No shield | --- | --- | Cgp | --- | 0.03 | pF |
| | | | | | | Cin | 3.0 | 5.0 | pF |
| | | | | | | Cout | 3.0 | 5.0 | pF |
| 1031 | Low-frequency vibration | - | Rp = 10,000 ohms | --- | --- | Ep | --- | 75 | mVac |
| --- | Reduced pressure (altitude) rating | - | pala = 8.2 ±0.5 mmHg | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 1 | | | | | | | | |
| 1266 | Total grid current (1) | 5 | Ec1 = -1.0 Vdc; Eb = 100 Vdc; Ec2 = 100 Vdc | 0.65 | II | Ic1 | 0 | -0.5 | µAdc |
| 1266 | Total grid current (2) | 5 | Ec1 = -45 Vdc | 0.65 | II | Ic1 | 0 | -0.5 | µAdc |
| 1256 | Electrode current (anode) | - | | 0.65 | II | Ib | 2.1 | 4.9 | mAdc |
| 1306 | Transconductance (3) | 5 | Ef = 1.0 Vdc | 0.65 | II | Sm | 1,500 | --- | µmhos |
| Ⓐ 1201 | Short and discontinuity detection | - | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | | |
| 1211 | Insulation of electrodes | - | | 4.0 | S3 | R | 500 | --- | Meg |
| 1116 | Lead fatigue | - | | 2.5 | Code F | --- | --- | --- | --- |
| Ⓐ 1119 | Glass-to-lead seal strain | - | | 6.5 | Code F | --- | --- | --- | --- |
| 1301 | Filament current | - | | --- | --- | If | 88 | 112 | mA |
| 1256 | Electrode current (screen) | - | | --- | --- | Ic2 | 0.4 | 1.4 | mAdc |
| 1306 | Transconductance (1) | - | | --- | --- | Sm | 1,800 | 3,200 | µmhos |
| 1306 | Transconductance (2) | - | Ec1 = -3.5 Vdc | --- | --- | Sm | --- | 75 | µmhos |
| 1316 | Amplification factor (triode) | 6 | | --- | --- | Mu | 36 | 49 | --- |
| 1016 | Moisture resistance | 7, 8 | | 6.5 | --- | --- | --- | --- | --- |
| Ⓐ 1105 | Permanence of marking | - | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | | |
| --- | Operation life | - | Group A; Ef = 1.25 Vac; Rg1 = 5.0 Meg | --- | --- | --- | --- | --- | --- |
| --- | Operation life-test end points (24 hours): | | | | | | | | |
| 1306 | Transconductance (1) | - | | --- | --- | Sm | 1,500 | --- | µmhos |
| 1266 | Total grid current (2) | - | | --- | --- | Ic1 | --- | -0.5 | µAdc |



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NOTES:

Ⓐ 1. The envelope is coated with a metallic shield which is connected to lead No. 3.

2. Grid No. 3 is composed of two separate deflector plates, one of which is connected to lead No. 3 and the other to lead No. 5.

Ⓐ 3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

Ⓐ 4. See MIL-E-1, appendix F.

5. This test shall be performed at the conclusion of the holding period.

6. Tie the anode and screen grid together as one element.

7. After completion of the test, the metallic coating shall be rubbed 12 times, manually, with as much pressure as possible exerted using a soft chamois cloth held firmly in the rubbing hand. Impairment of the coating, as evidenced by blisters, cracks, or peeling, shall be cause for rejection.

Ⓐ 8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2847)



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